

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 25

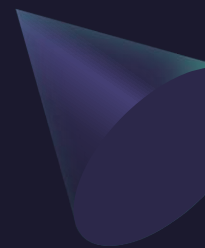


Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.5, 16.6 (curl, divergence, parametric surfaces)
- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!



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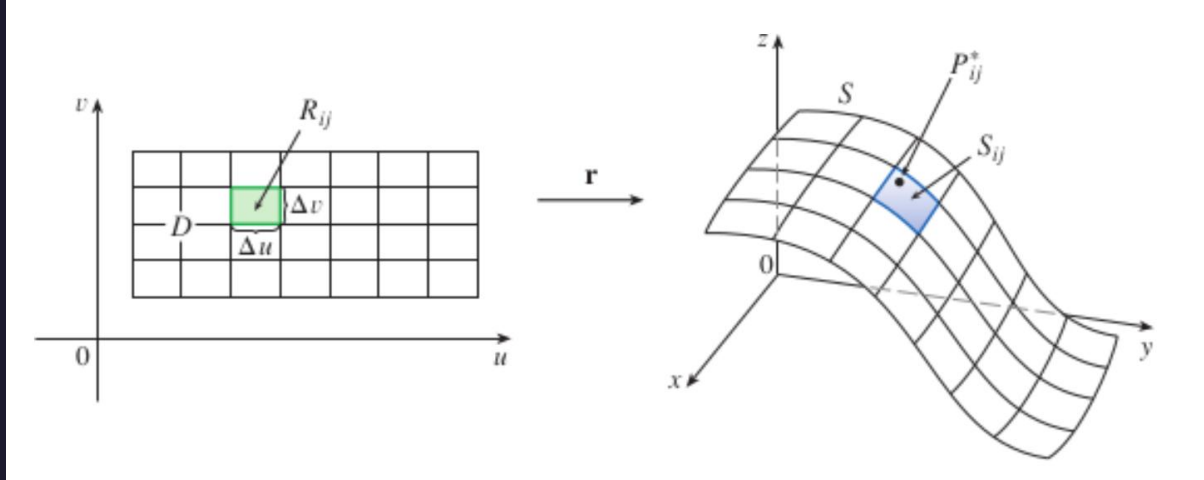
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Surface Integrals



Integrals on Parametrized Surfaces

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



$$\iint_S f(x, y, z) \, dS = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \, \Delta S_{ij}$$

$$\iint_S f(x, y, z) \, dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

Integrals on Graphs of Functions

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

Orientation

- ▶ For surface integrals of a vector field $\vec{F}$ over a surface S , the orientation of the surface must be specified, just like the line integrals. The orientation of a surface is specified via its normal vectors at any point.
- ▶ For a closed surface S , we define **the positive orientation** of S to mean that all normal vectors to S point outward (with negative meaning normal vectors point inward instead). If it is impossible to have all normal vectors always point either outward or inward, then the surface S is non-orientable.

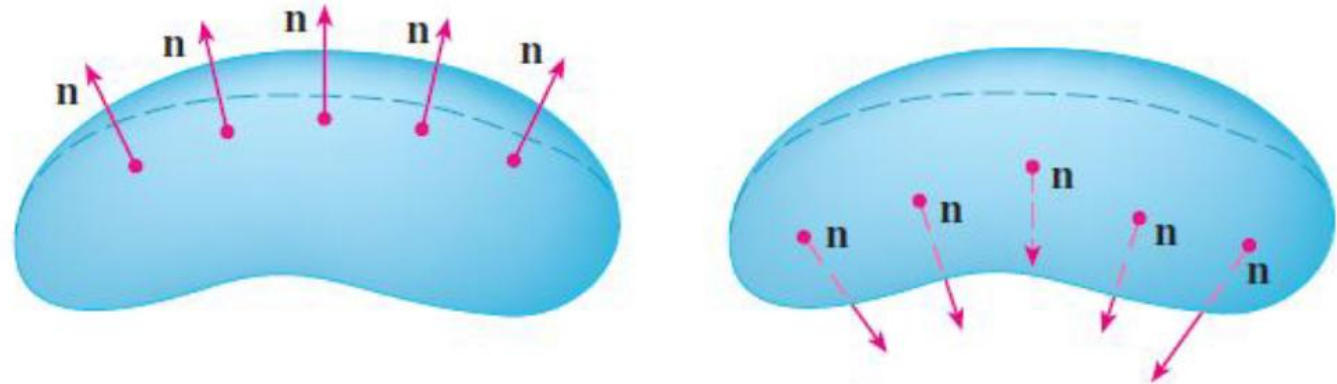


FIGURE 7

The two orientations
of an orientable surface

- ▶ To define a surface integral of a vector field, we only want to count the portion of $\vec{F}$ that is normal to the surface S at any point. Hence, the dot product again plays a role in the computation.
- ▶ Then the **surface integral** of $\vec{F}$ over the surface S is defined as

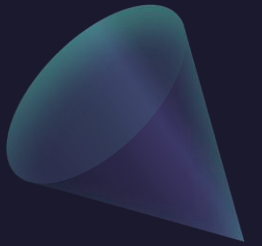
$$\begin{aligned}\iint_S \vec{F} \cdot d\vec{S} &= \iint_S \vec{F} \cdot \vec{n} \, dS \\ &= \iint_S \vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \, dS \\ &= \iint_D \left[\vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \right] |\vec{r}_u \times \vec{r}_v| \, dA \\ &= \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, dA.\end{aligned}$$

This is often called the **flux integral** as it measures the flux of $\vec{F}$ across the surface S .

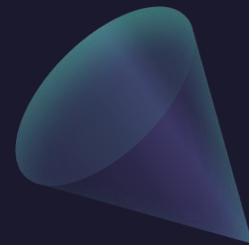
- ▶ Recall and compare it with the line integral of vector fields:

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt.$$

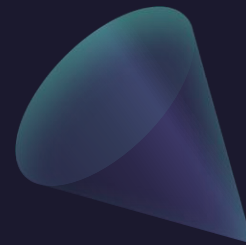
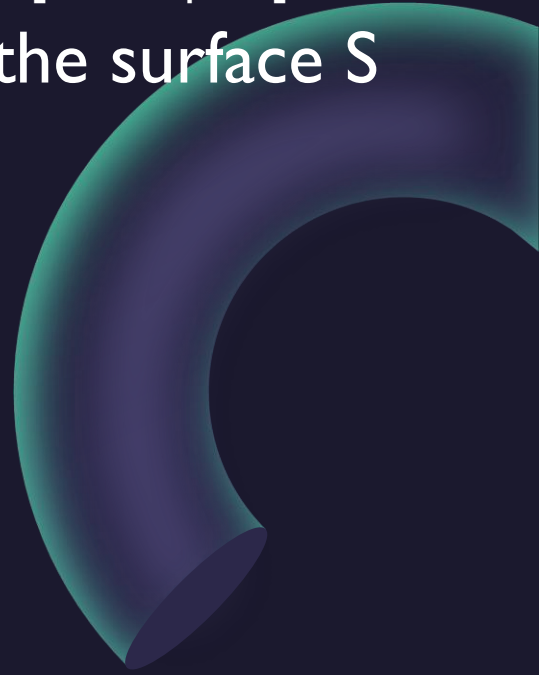
Orientation



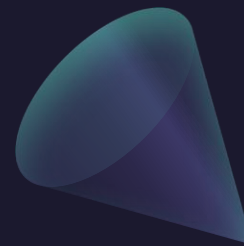
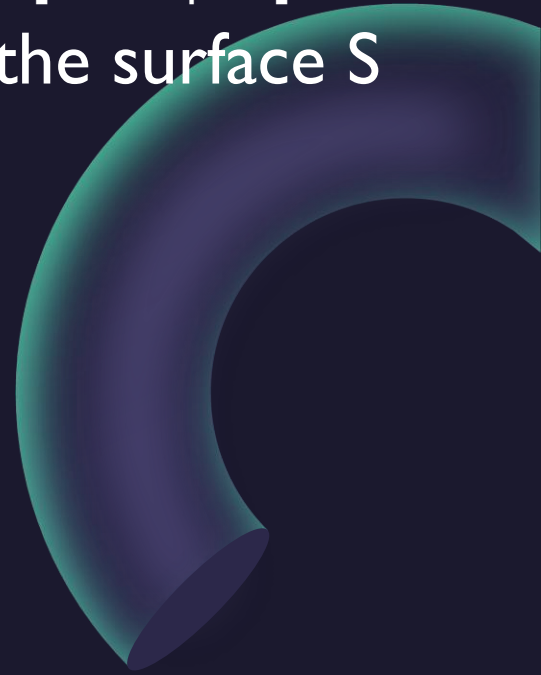
Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$.
(Assume S is positively oriented.)



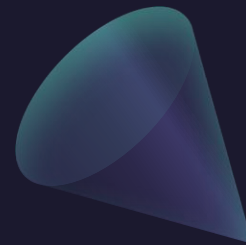
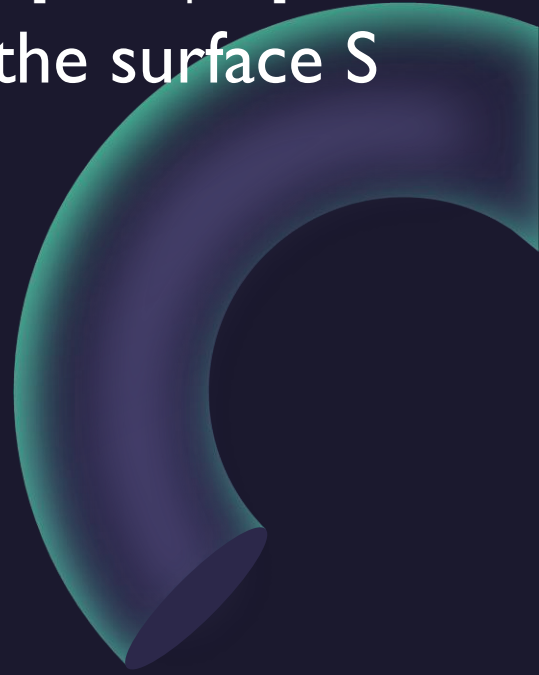
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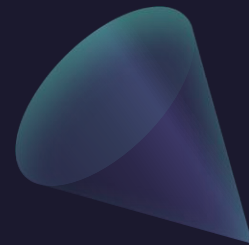
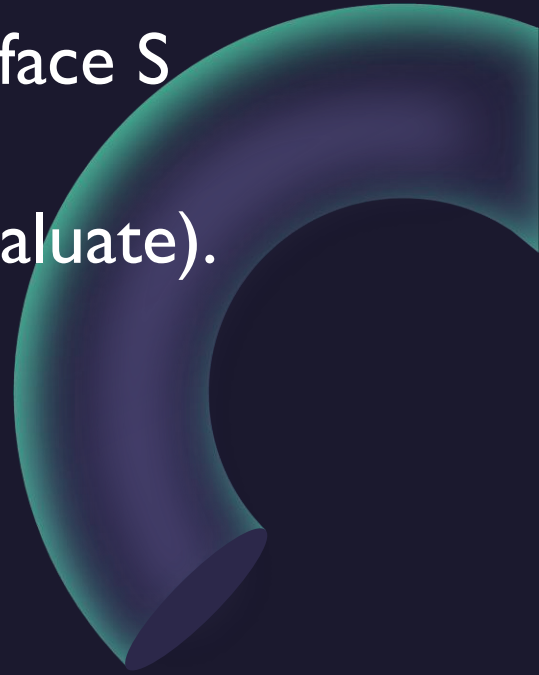


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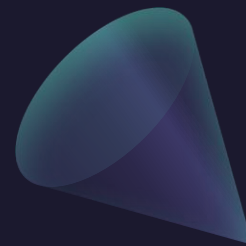
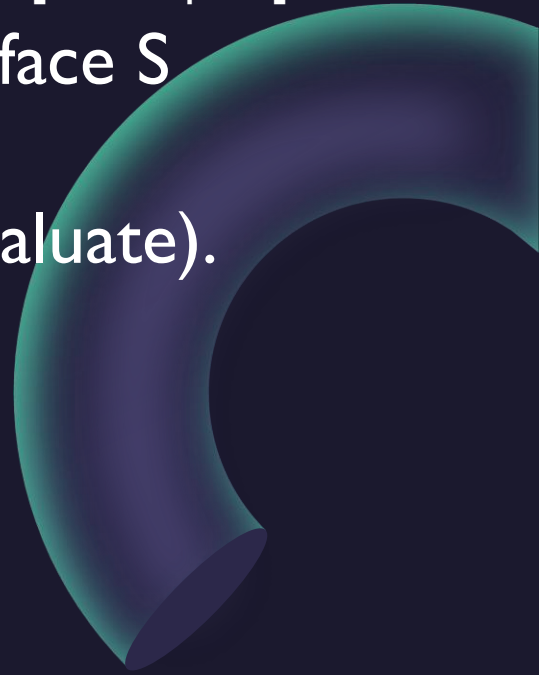
Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \operatorname{curl}(F) \cdot dS$$



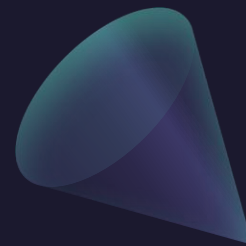
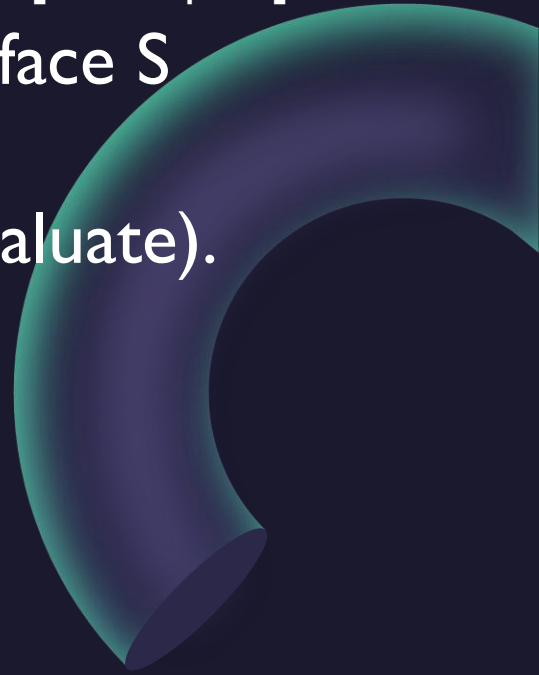
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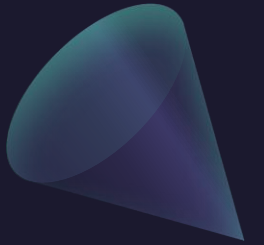


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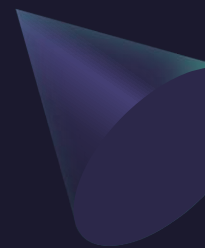


Questions?





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Stokes' Theorem



Recall: Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P \, dx + Q \, dy$$

$$(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Then Green's Theorem says:

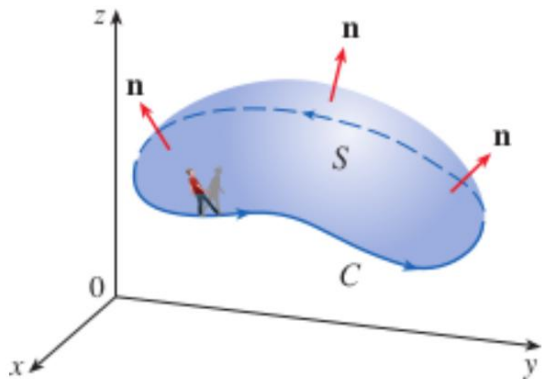
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$$

Stokes' Theorem

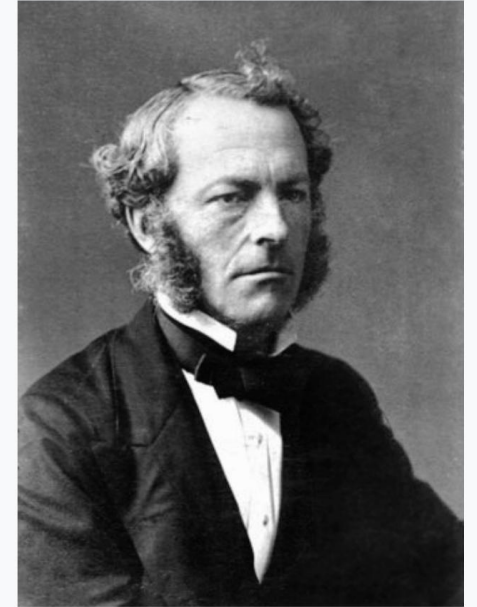
Stokes' Theorem

Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$



Sir
George Stokes
FRS



Stokes, 1860s

35th **President of the Royal Society**

In office
1885–1890

Preceded by [Thomas Huxley](#)

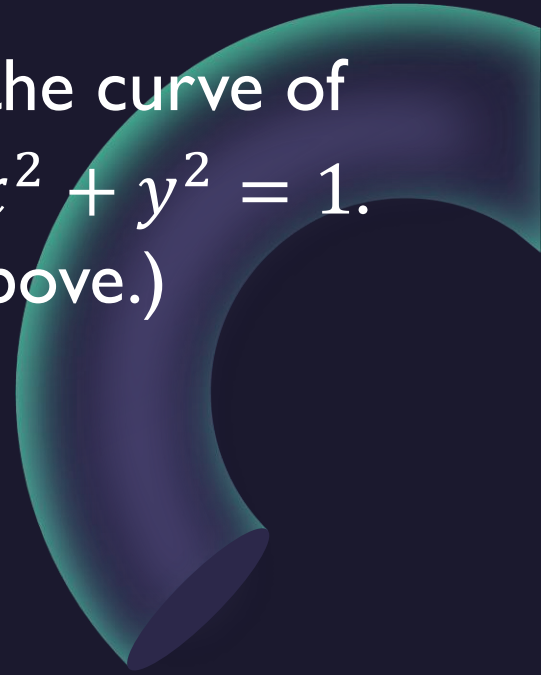
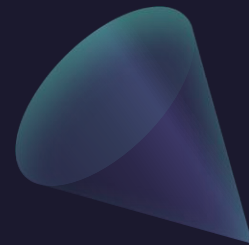
Succeeded by [Lord Kelvin](#)

Personal details

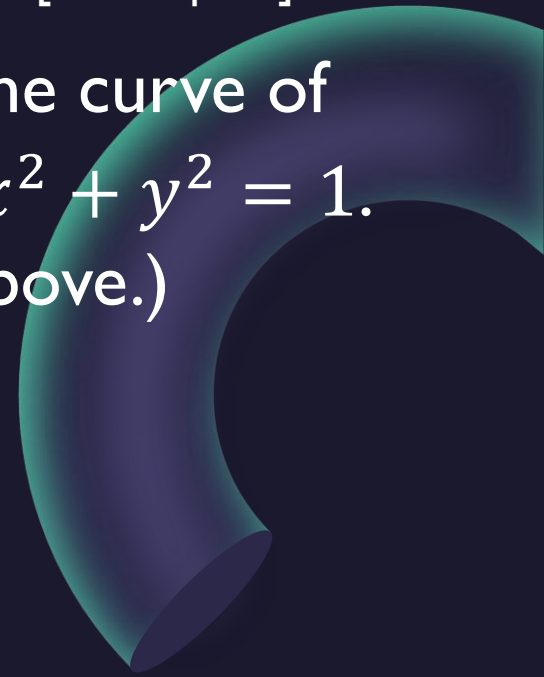
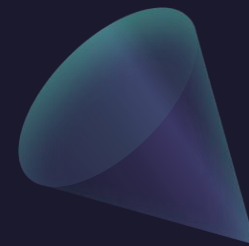
Born George Gabriel Stokes
13 August 1819
[Skreen](#), Ireland, United Kingdom of Great Britain and Ireland

Died 1 February 1903 (aged 83)
[Cambridge](#), England, United Kingdom of Great Britain and Ireland

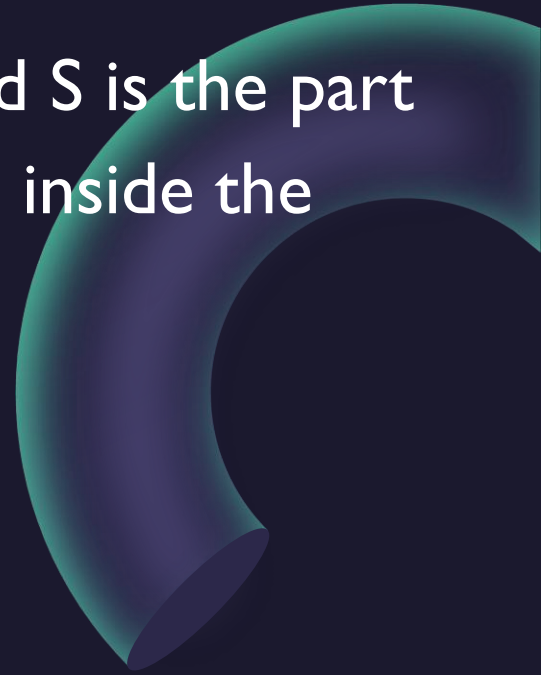
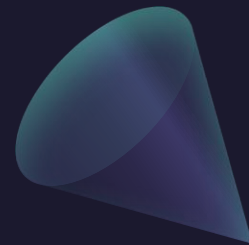
Example: Evaluate $\int_C F \cdot dr$ where $F = (-y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (C is oriented counterclockwise when viewed from above.)



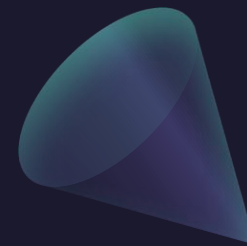
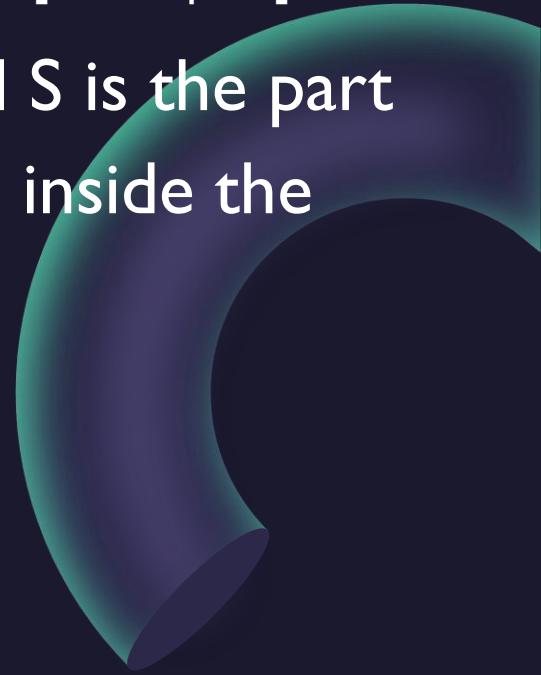
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Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

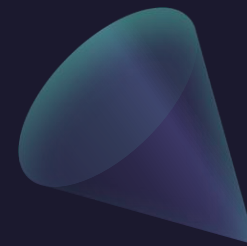
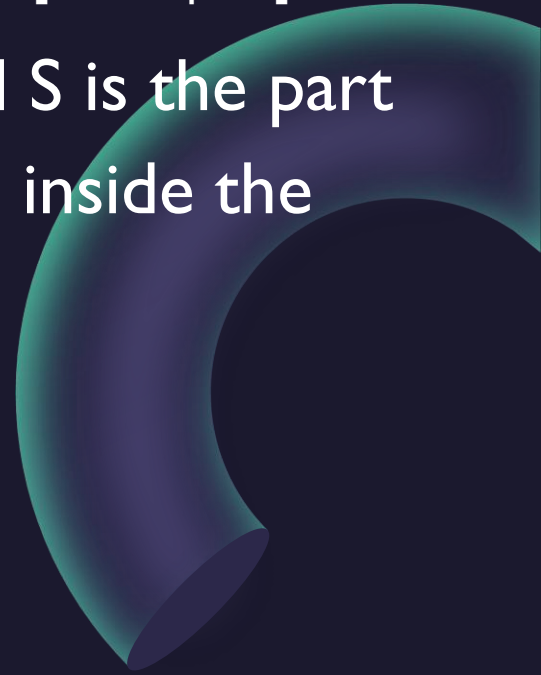


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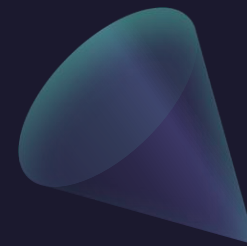
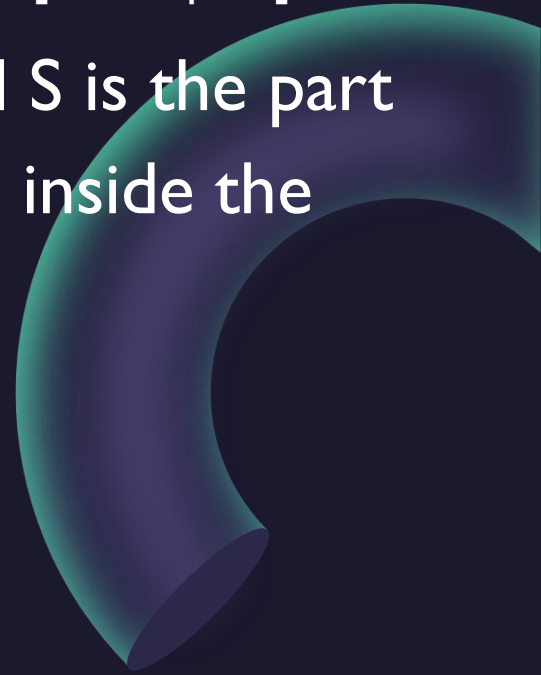
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Alternative solution: Change S to a disk!

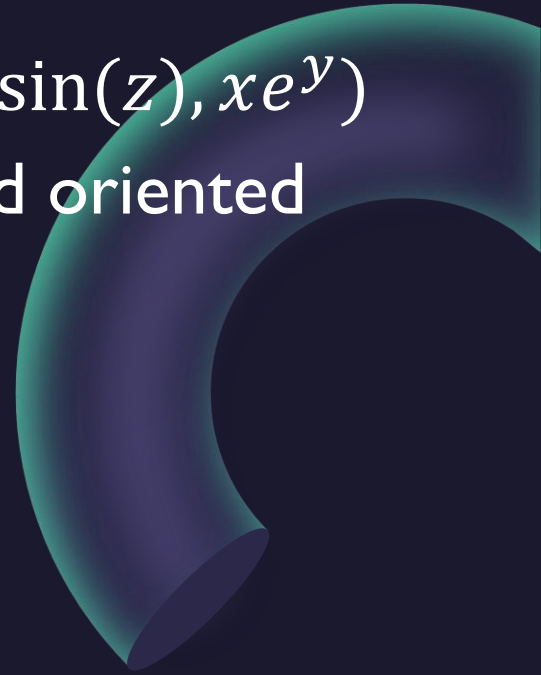
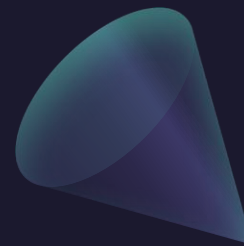


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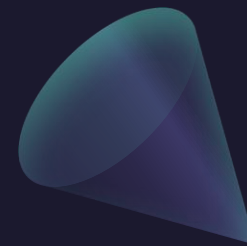
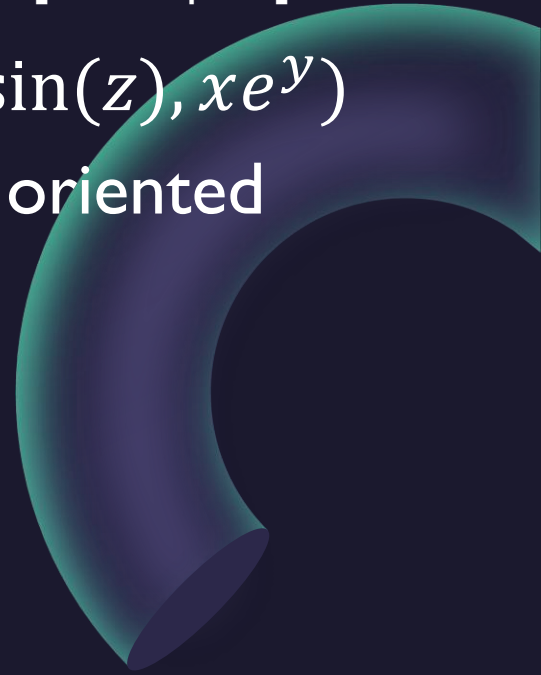
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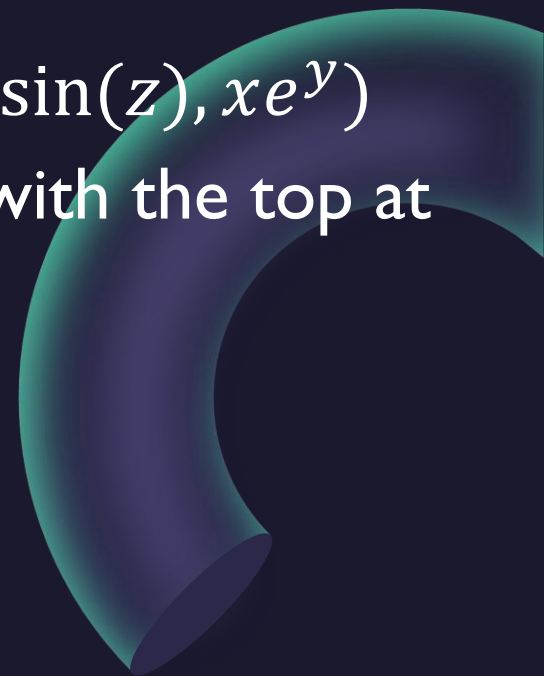
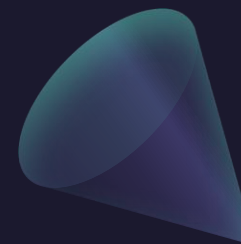
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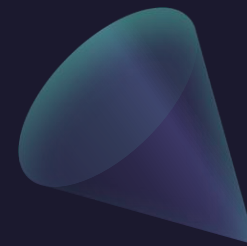
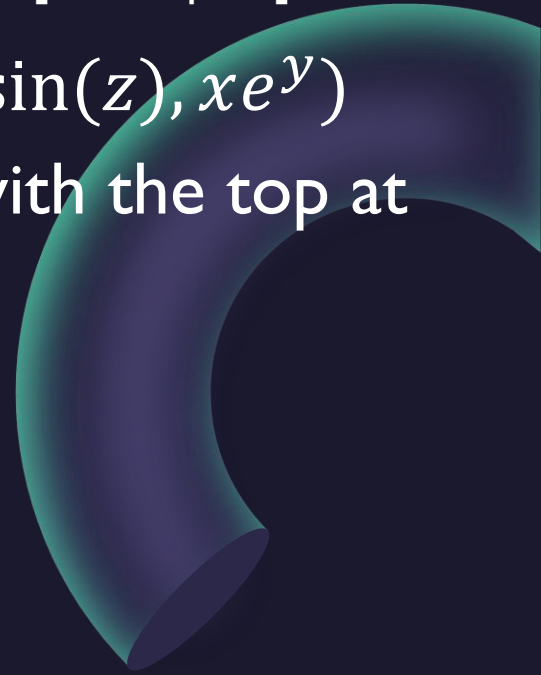
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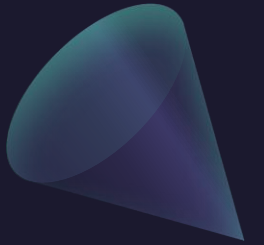
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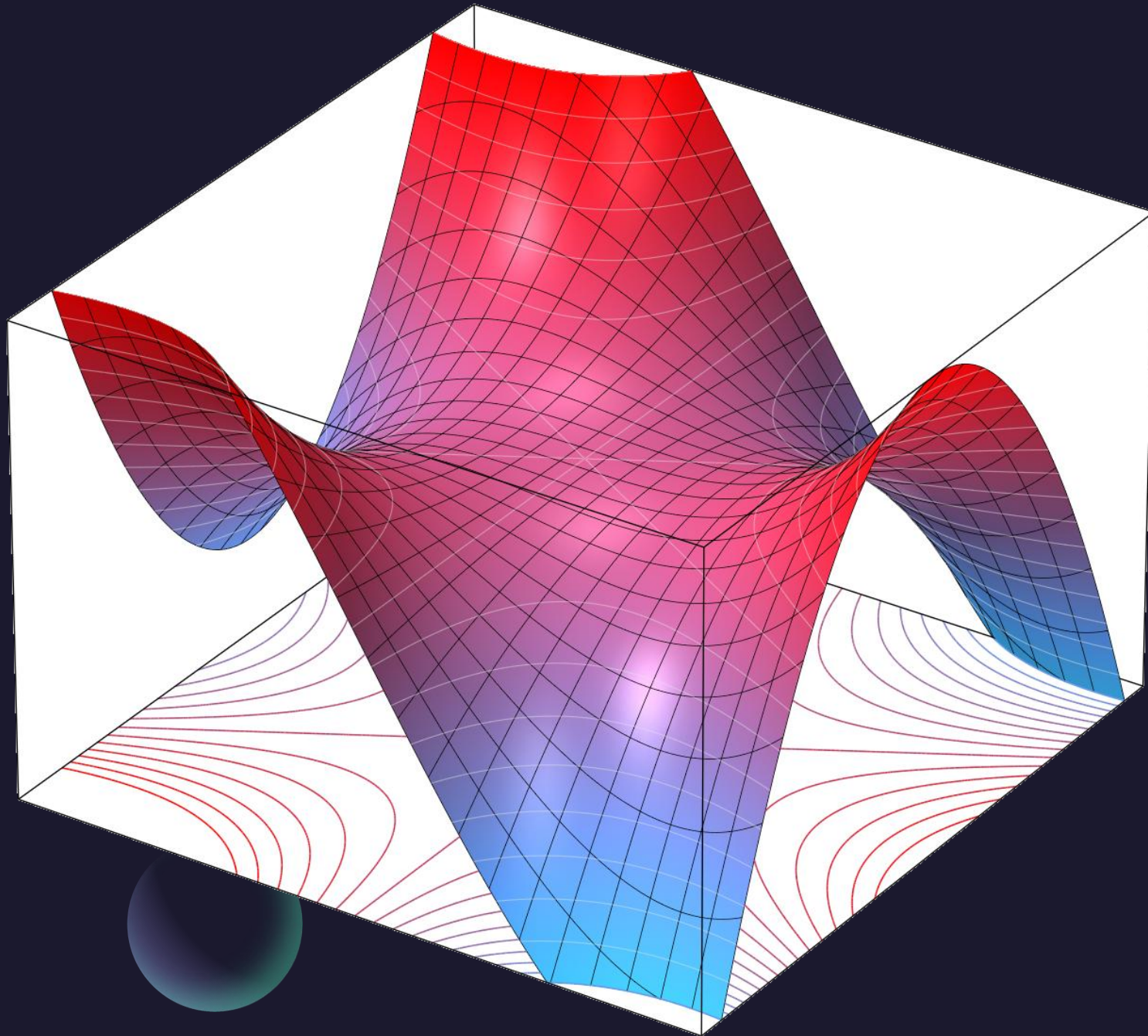


Questions?



Thank you

Until next time.



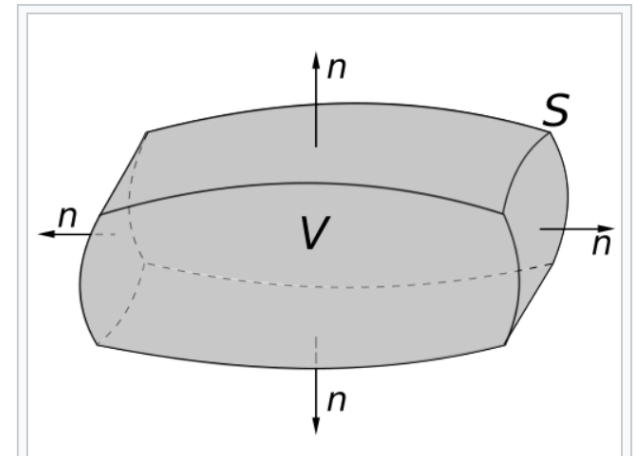
The Divergence Theorem (Gauss's Theorem)

Mathematical statement [\[edit\]](#)

Suppose V is a [subset](#) of $\mathbb{R}^n$ (in the case of $n = 3$, V represents a volume in [three-dimensional space](#)) which is [compact](#) and has a [piecewise smooth boundary](#) S (also indicated with $\partial V = S$). If $\mathbf{F}$ is a continuously differentiable vector field defined on a [neighborhood](#) of V , then:[\[4\]](#)[\[5\]](#)

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \oiint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) \, dS.$$

The left side is a [volume integral](#) over the volume V , and the right side is the [surface integral](#) over the boundary of the volume V . The closed, measurable set ∂V is oriented by outward-pointing [normals](#), and $\hat{\mathbf{n}}$ is the outward pointing unit normal at almost each point on the boundary ∂V . ($d\mathbf{S}$ may be used as a shorthand for $\mathbf{n}dS$.) In terms of the intuitive description above, the left-hand side of the equation represents the total of the sources in the volume V , and the right-hand side represents the total flow across the boundary S .



A region V bounded by the surface $S = \partial V$ with the surface normal n